

Financial versus Strategic Buyers

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Abstract

This article introduces the impact of debt misvaluation on merger and acquisition activity. We show the potential for debt misvaluation to help explain the shifting dominance of financial acquirers (private equity firms) relative to strategic acquirers (operating companies). Fundamental differences in governance and project coinsurance between the two types of acquirer would interact with debt misvaluation, resulting in variation in how assets are owned that depends on debt market conditions. We find support for our theory in merger data using a novel measure of debt misvaluation.

I. Introduction

Mergers and acquisitions (M&A) occur in waves of activity with recent troughs, for example, of only a few thousand deals in 2003 and peaks of over 10,000 deals in 1999 and 2006.¹ Within this oscillation of activity there is another shifting pattern: The percentage of so-called financial sponsors (private equity (PE) firms) versus strategic buyers (operating companies) seems to ebb and flow.

Figure 1 examines the financial sponsor versus strategic proportion of M&A activity of all public targets with values of less than \$1 billion recorded in the SDC Platinum database from 1984 to 2010.² It is immediately clear that the fraction of

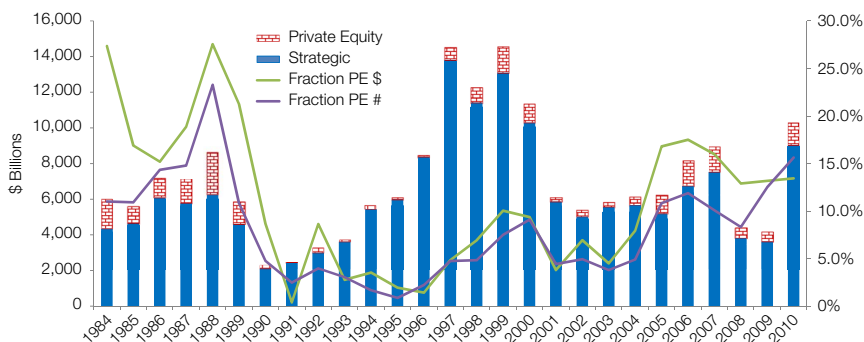
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¹U.S. merger activity as reported by SDC Platinum database. See Harford (2005), Rhodes-Kropf, Robinson, and Viswanathan (2005), Mitchell and Mulherin (1996), Andrade, Mitchell, and Stafford (2001), and Holmstrom and Kaplan (2001), for empirical evidence on merger waves.

²PE firms are limited in the size of checks they can write to buy a firm by the amount they have under management and covenants with their investors, called limited partners (LPs). Both strategic and

FIGURE 1
U.S. M&A Volume (\$10M–\$1B) by Type of Buyer

Figure 1 plots U.S. volume (in dollars) of M&As by private equity (PE) (financial) and strategic buyers (left axis). The figure also plots the corresponding share of PE acquisitions based on both deal count (Fraction PE #) and transaction value (Fraction PE \$) (right axis). The sample is restricted to all M&A transactions involving a public target with transaction values greater than \$10 million and lower than \$1 billion, recorded in the SDC Platinum database from 1984 to 2010.



total deal value acquired by financial sponsors has varied dramatically over the last 25 years. This pattern is true across many industries.

Any particular transaction has many factors that drive the ultimate acquirer's willingness-to-pay (WTP). Many theories propose reasons that particular firms or industries may be ripe for acquisition activity (see Jensen (1986), Maksimovic and Phillips (2001), Gorton, Kahl, and Rosen (2009), Jovanovic and Rousseau (2002), Mitchell and Mulherin (1996), Lehn, Netter, and Poulsen (1990), Mulherin and Boone (2000), and Harford (2005)). However, the broad pattern of financial sponsor activity suggests a broad economic explanation for the coordination.

What drives either financial or strategic buyers to have a more dominant position in M&A activity at different points in time? This question is important not only because the economic magnitude of this activity is so large but also because the balance of power between financial and strategic acquirers changes the ownership structure of assets and alters the incentives and governance mechanisms that impact how firms are managed.

One potential broad economic mechanism that would imply a shifting WTP by strategic acquirers stems from Rhodes-Kropf and Viswanathan (2004) and Shleifer and Vishny (2003), who suggest that strategic acquirers with overvalued stock will bid more, and overvalued targets are more willing to accept takeover offers.³ But clearly, financial buyers who must pay in cash should avoid overvalued targets. This implies that patterns of financial versus strategic activity could be driven by the same phenomenon. However, a quick look at Figure 1 suggests

financial buyers can reasonably acquire public targets with values of less than \$1 billion. Increasing the target size cutoff dampens the percentage of PE activity in every period, but the increases and decreases in activity are still evident. We also removed deals of less than \$10 million as a standard screen. See Fruhan (2010) for more on the role of PE in acquisitions and the shifting pattern across time. In the regressions that follow, we do not cut the data by the size of the transaction.

³Support for the misvaluation theory has been found by Rhodes-Kropf et al. (2005), Ang and Cheng (2006), Dong, Hirshleifer, Richardson, and Teoh (2006), and others.

that something else must explain financial sponsor activity as the local peaks of financial sponsor activity *relative* to strategic activity correspond with stock market peaks like those in the late 1990s and 2006–2007, with dips in the early 1990s and 2001 recessions.

An alternative notion comes from Harford (2005), who shows that interest rates are significantly inversely correlated with merger activity. Harford proposes no formal theory of merger activity, but argues that a low spread is a proxy for overall liquidity or ease of financing. Although easy access to financing should impact merger activity, it is not clear how this would differentially affect acquirer types.

In this article, we combine ideas from previous work and introduce the possibility of misvalued debt markets. We demonstrate how misvalued debt can both fuel M&A activity and alter the balance between financial and strategic buyers.

Much recent research has focused on private equity (see Kaplan and Stromberg (2009) for an overview). However, little research directly considers the competition between financial and strategic buyers. Recent papers by Barger, Schlingemann, Stulz, and Zutter (2008), Hege, Lovo, Slovin, and Sushka (2012), and Dittmar, Li, and Nain (2012) focus on bidding behavior and target premiums between strategic and financial acquirers. Gorbunov and Malenko (2014) consider the bidding behavior of strategic versus financial bidders, focusing on how a strategic's search for synergies causes different bidding behavior than a financial buyer's search for undervalued assets. However, this research does not offer any broad insights into the rising and falling tides of PE activity. Axelson, Jenkinson, Stromberg, and Weisbach (2013) complete an empirical analysis of PE transactions and Axelson, Stromberg, and Weisbach (2009) propose a theory of PE capital structure, which together suggest that the PE phenomenon is related to debt market conditions.⁴ Haddad, Loualiche, and Plosser (2011) offer an alternative view, arguing that more leveraged buyouts (LBOs) should occur when risk-free rates are high and the risk premium is low due to the benefits and cost of concentrated ownership. Shivdasani and Wang (2011) report that structured credit fueled the most recent buyout boom, and Holmstrom and Kaplan (2001) document and discuss the LBO wave in the late 1980s. This work does not consider PE relative to strategic activity through the different merger waves, nor does any of this past work consider debt market misvaluation. Much like the work in M&A that has shown effects due both to misvaluation and changing economic conditions, it is likely that buyout activity is affected both by fundamentals and misvaluation.

While it seems reasonable to assume that if equity markets can be misvalued, then so can debt markets, it is much less obvious that “cheap” debt should lead to more acquisition activity. After all, the targets can also access cheap debt and so are more valuable as stand-alone entities when debt is cheap. On top of this, it is not clear how debt misvaluation should alter the interplay between financial and strategic buyers. Just believing that debt markets are overvalued does not imply a benefit to one type of buyer. If both types of acquirers find a misvalued debt market, cannot both take advantage of it? Since it is not *ex ante* obvious

⁴Müller and Panunzi (2004), Morellec and Zhdanov (2008), and Ivashina and Kovner (2011) also examine the leverage of financial buyers.

what misvalued debt might do to the market for acquisitions or how it would differentially impact the participants, our model provides important insights and understanding. We also provide some evidence to support our main theoretical implications in the data to help show the strength of the theory.

While each type of buyer and targets can equally access the debt market, fundamental differences alter the benefit to each: i) Strategic buyers have a current project (or projects) they are considering combining with the target, while financial sponsors evaluate the target as a stand-alone project. ii) Financial buyers have a different corporate governance structure (monitoring technology) than strategic buyers. We interact misvalued debt with these fundamental differences to develop both a coinsurance effect and a monitoring effect of misvalued debt.

Coinsurance arises anytime less-than-perfectly correlated projects are combined. This idea was first proposed by Lewellen (1971), extended by Higgins and Schall (1975) and Galai and Masulis (1976), and has been repeatedly considered in the financial literature, both empirically and theoretically.⁵ We build on this work to examine how coinsurance interacts with the potential misvaluation of debt claims even when agents are risk neutral.

We show that the interaction of overvaluation and coinsurance works against strategic buyers. This occurs because the investors' overvaluation of projects/firms causes them to also misperceive the value of coinsurance. A strategic buyer combines projects, effectively reducing the valuation mistake being made by investors because mistakes partially offset each other. That is, they underestimate the default probability, but that makes them underestimate the benefit of project coinsurance as well. Therefore, strategic buyers are less able than financial buyers to exploit investors' misperceptions. In sum, even though both strategic and financial buyers would like to take advantage of interest rates that are too low and avoid borrowing when interest rates are too high, they are differentially impacted by the errors and are willing to pay *relatively* more or less depending on the sign of the error made on interest rates.

A debt contract is optimal in our setting because of managerial moral hazard, which also creates a role for costly monitoring. A fundamental advantage of PE ownership is often thought to be better oversight and governance. Thus, if misvaluation interacts with the moral hazard problem, then PE and strategic buyers will be differentially affected. We show that overvaluation increases the moral hazard problem and enhances the importance of better governance to eliminate misbehavior. This allows PE buyers to both increase leverage and pay more.

Thus, misvaluation interacts with both the effect of governance and the perceived value of coinsurance, creating a relative advantage to PE buyers in overvalued debt markets.

Note that although the possibility of overvalued debt may help financial buyers win the target, overvalued debt may not help financial buyers' returns. Overvalued debt increases all financial buyers' WTP, but competition may cause the

⁵For example, see Kim and McConnell (1977) for an early empirical examination of coinsurance on debt prices after mergers, while Leland (2007) completes an in-depth theoretic examination, and Faure-Grimaud and Inderst (2005) consider the effect of uncorrelated projects in the context of conglomerates and governance.

gains to go to the target. Since PE firms are more likely to win in overvalued credit markets, we show that they should use more leverage and pay higher prices. Axelson et al. (2013) find support for this idea and report that credit market conditions affect the prices paid and are the main driver of the quantity of debt used in buyouts. Furthermore, Axelson et al. (2013) find that highly levered transactions are associated with lower fund returns. Also, Hege et al. (2012) report that sellers of assets to PE buyers earn positive returns significantly greater than in sales to public operating firms.

The joint presence of moral hazard and misvaluation yields interesting insights and allows us to contribute at a methodological level to the literature by analyzing an agency model with asymmetric information between investors and managers.⁶ Moreover, our model also has a number of interesting empirical implications. First and foremost, the possibility of misvalued debt not only changes the likelihood of an acquisition, it also changes the type of buyer and the way the assets are owned. These predictions have empirical content because, although debt market mispricing may be difficult to determine in real time, by looking backward we should be able to find times when credit was particularly misvalued and see whether debt overvaluation corresponds with increased M&A activity and increased PE activity relative to strategic buyers.

We take our predictions to the data to show some evidence on the effect of misvalued debt. We introduce a novel measure of debt market overvaluation and find that it strongly correlates with the ratio of PE to strategic merger activity. Moreover, debt market overvaluation drives out any relationship between the ratio of PE to strategic merger activity and the high-yield credit spread.

Our theory also predicts that PE firms will tend to dominate strategic buyers when debt markets are overvalued, but the relative dominance of PE firms to strategic buyers should be even greater if the strategic acquirer is a conglomerate. We also find support for this prediction and some additional predictions in the data.

Section II describes the model. In Section III, we derive the maximum WTP depending on the organizational form in which the firm is organized. The main results of the paper are in Sections IV, V, and VI, in which we explain the model's implication regarding misvaluation and coinsurance, misvaluation and moral hazard, and the effect of misvaluation on aggregate acquisition activity, respectively. In Section VII, we discuss the robustness of the model as well as some empirical implications, and Section VIII presents some evidence. Section IX concludes. The online Supplementary Material contains the proofs of all propositions not included in the article.

II. The Model

The basic setup comes from the workhorse model of Holmstrom and Tirole (1997) with some interesting additions. We use this model because it was

⁶Our paper contributes to the burgeoning literature on the effect of differential beliefs on contracting such as Landier and Thesmar (2009), Gervais, Heaton, and Odean (2011), and Goel and Thakor (2008). While these papers look at managers who are overoptimistic, we assume the manager is correctly informed but the investor mistakes (possibly optimism) interact with the moral hazard problem.

originally used to model lending in an environment with moral hazard and governance through monitoring. Thus, although the model makes some assumptions to drive tractability, we can use it to examine the effects of both leverage and governance. Both effects are central to the PE model (see Kaplan and Stromberg (2009)).

The economy consists of three types of agents: managers, PE partners, and investors. They differ in both their abilities to generate returns and their information sets, in a way that will be clear shortly. All agents are risk neutral. We differ from Holmstrom and Tirole (1997) in that we add potential misvaluation and focus on the highest WTP for a project as well as whether the project can obtain financing.

A. Managers and Private Equity Partners

A project for sale has a current manager, who owns the project, and two potential buyers: a PE firm (who joins with a manager) and a manager with a current project (a merger or strategic acquirer). Regardless of whether the project is purchased, it requires an investment I in period 1 to realize its return in period 2. In period 2, the investment generates a verifiable return equaling either 0 (failure) or a positive return (success). A stand-alone project returns R , a project with a PE partner who monitors the project returns R^{PE} , and a project together with another project returns R^s per project.

The probability that the project succeeds is p_H if the manager works hard or $p_L \equiv \theta p_H$ if the manager shirks, where $\theta < 1$. Let us define $\Delta p \equiv p_H - p_L = (1 - \theta)p_H > 0$. Projects are run by managers who receive private benefits of b or B if they shirk, where $0 < b < B$. Thus, without proper incentives, managers will shirk and create lower expected return projects with higher private benefits.

The potential benefit of including a PE investor is that PE firms can monitor the project and reduce the private benefit from B to b . However, PE firms must pay a cost, $c > 0$, to monitor and will therefore monitor only if they have the incentives to do so. We assume that $b + c - B > 0$, so that there is a tradeoff to engaging with a PE firm and not all firms should do so. Let $\bar{V}^{\text{PE}}$ represent the maximum a PE firm is willing to pay for the project and $\bar{V}^s$ represent the maximum a manager with another project (strategic acquirer) is willing to pay. The stand-alone manager values the project at $\bar{V}$, which is just the highest amount they could extract from the firm if it is not sold.

The price paid by a buyer is for the right to invest I in the project. Therefore, the total amount needed for the project is $\bar{V} + I$. The buyer may pay for the project and the needed investment with either the cash they possess or by raising money from investors.

Stand-alone and strategic managers have capital A_m and PE investors may choose to invest capital A_{PE} . In order to focus on the interesting case when outside investors are needed, we will assume that $A_m + A_{\text{PE}} < I$. Initially, and to facilitate the comparison between different organizational forms, we assume $A_{\text{PE}} = 0$; then, in Section VII.A, we consider how different amounts of internal funds alter strategic and financial buyers' WTP. We also assume that there are infinitely many investors who do not monitor and demand an expected return of γ .

Table 1 summarizes the three possible organizational forms.

TABLE 1
Summary of Parameters and Assumptions

In Table 1, we list the main parameters of the model. The first three columns break the parameters down by organizational form (stand-alone, PE acquisition, and strategic acquisition, respectively). The "Assumption" column describes any assumption made with regard to those parameters.

	Stand-Alone	PE Acquisition	Strategic Acquisition	Assumption
Success return	R	R^{PE}	$2R^s$	—
Maximum value	$\bar{V}$	$\bar{V}^{\text{PE}}$	$\bar{V}^s$	—
Private benefits	B	b	$2B$	$B > b$
Monitoring cost	—	c	—	$b + c - B > 0$
Internal funds	A_m	$A_m + A_{\text{PE}}$	$2A_m$	$I > A_m + A_{\text{PE}}$

B. Uninformed Investors

The most interesting addition to the standard modeling assumptions is the potential for investors to not know and thus estimate with error or misperceive the probability of success and failure. We assume that all managers (including acquiring managers) and PE investors know p_H and p_L . However, investors do not know the true probabilities and instead use the probabilities p'_H and p'_L in assessing expected values.

A difference between the probabilities used by managers and those used by investors could arise fully rationally due to asymmetric information, as in Myers and Majluf (1984). Or, any biases, irrationality, or limited cognitive ability and limits to arbitrage could also result in an equilibrium misperception in the probability of success (see Barberis and Thaler (2003), Hirshleifer (2001), and Shleifer (2000) for summaries). In this paper, we take no stand on the source of the mistake, noting only that it is possible for investors to be mistaken. This is discussed further in Section VII.B.

Uninformed investors still require a return γ and have probability beliefs such that projects are economically viable only if the manager works hard, that is:

$$(1) \quad p'_H R - \gamma I > 0 > p'_L R - \gamma I + B.$$

If debt markets can be under- or overvalued, then the obvious conclusion is that firms should tend to issue more debt when it is overvalued. Furthermore, it would seem that all types of buyers and the stand-alone firm could take advantage of it equally, but we will see that this is not the case. Instead, valuation bubbles in the debt market can lead to overall increases in acquisition activity but also to waves of dominance of one type of buyer over another.

III. Organizational Forms

This section will determine the highest amount the strategic buyer and the PE firm would be willing to offer while simultaneously determining the reservation price of the stand-alone firm. With these benchmarks established, the following sections will examine the drivers that provide each type of organizational form a higher WTP and thus an advantage in the takeover market.

A. Strategic Acquisition

We first consider a manager who already has a project and is trying to buy a second project. We will call this manager a strategic acquirer.

Our definition of a firm is a pair consisting of a project and a manager. Therefore, after a strategic acquisition, the new entity will have two managers, each with a project and each with access to cash in the amount of A_m . Each project requires an investment of I and generates a return R^s with the real and perceived probabilities as described previously. Thus, we assume the projects are uncorrelated.⁷ The payoffs of all claims are based on the outcome of both projects. Thus, here we rule out project financing, as this would be the same as a manager with a single project, which we analyze in the next subsection.⁸

With two projects, an optimal contract requires managers to invest $2A_m$ and the uninformed investors to invest the balance of $2(\bar{V}^s + I - A_m)$. An optimal contract then pays the two managers nothing if either or both projects fail.⁹ If both projects succeed, the contract divides the payoff $2R^s$ into $2R_m^s > 0$ for both managers (each receives R_m^s) and $R_u^s > 0$ for the uninformed investor, where $2R_m^s + R_u^s = 2R^s$. Thus, the contract is debt because it does not give investors the same share of any payoff but rather pays investors all of the cash flows in the worst states and only part of the payoff in the best state.

Given that managers and projects are symmetric, incentives will always be such that both managers will work hard or both will shirk.¹⁰ Thus, given that projects are economically viable only if managers work hard, uninformed investors will invest only if they believe managers will work hard. A manager, however, only gets paid if both projects pay off, thus investors need to believe that $p_H^2 R_m^s \geq p'_H p'_L R_m^s + B$.¹¹ Therefore, the investor's beliefs about the incentive-compatibility (IC) constraint requires that the manager be paid at least

$$(2) \quad \text{Investor's view of IC:} \quad R_m^s \geq B / p'_H \Delta p'.$$

However, since $p'_H \Delta p' \neq p_H \Delta p$, the manager will not actually work hard unless $R_m^s \geq B / p_H \Delta p$. We must account for this when we consider the manager's individual rationality constraint.

If condition (2) holds, then uninformed investors will invest $2(\bar{V}^s + I - A_m)$ if they expect to earn γ on this investment. Thus, $p_H^2 R_u^s + 2p'_H(1 - p'_H)R^s = 2\gamma(\bar{V}^s + I - A_m)$. Note that this equation uses the fact that in case only one of the projects is successful the payoff to the investor is the entire cash flow available,

⁷Section IV.A examines the effect of correlated projects.

⁸This is equivalent to assuming that the returns from each project cannot be verifiably attached to that project.

⁹The Supplementary Material proves this formally, following Holmstrom and Tirole (1997) and Tirole (2005).

¹⁰If the first manager working hard means the second manager wants to shirk, then $p_H^2 R_m^s < p_H p_L R_m^s + B$. But this implies $p_H p_L R_m^s < p_L^2 R_m^s + B$ because $p_L / p_H < 1$, which means that if the second manager shirks then the first will as well. So it is a Nash equilibrium for both to shirk. If the first manager shirking means the second manager wants to work hard, then $p_H p_L R_m^s \geq p_L^2 R_m^s + B$. But this implies $p_H^2 R_m^s \geq p_H p_L R_m^s + B$ because $p_H / p_L > 1$, which means that if the second manager chooses to work hard then so will the first. So it is then a Nash equilibrium for both to work hard.

¹¹We assume investors focus on the Pareto-dominating equilibrium, in which both managers work hard as long as it is incentive compatible.

R^s , thus investors retain a debt-like priority. Hence the only unknown variable is R_u^s .

Given the previous equations, both managers will work hard only if

$$(3) \quad \text{Manager's IC:} \quad R_m^s = R^s - \gamma(\bar{V}^s + I - A_m)/p_H^2 \\ + (1 - p_H')R^s/p_H' \geq B/p_H \Delta p.$$

Investors will provide debt for the project only if they believe managers will work hard.

That there is both a true IC constraint and a different perceived IC constraint is a novel aspect of our model. Only if the *perceived* IC constraint (equation (2)) holds will investors invest, but only if the actual IC constraint (equation (3)) holds will managers work hard.

The manager's expected return must also be greater than γA_m , otherwise she would rather invest A_m elsewhere. We assume an inelastic supply of projects that earn γ . Therefore, the manager's individual rationality constraint (IR) is

$$\text{Manager's IR} \quad \begin{cases} R_m^s \geq \gamma A_m/p_H^2 & \text{if manager's IC holds} \\ R_m^s \geq (\gamma A_m - B)/p_H p_L & \text{if manager's IC does not hold.} \end{cases}$$

This is different from a standard model without misvaluation because in a standard model the manager's IC always holds in equilibrium. However, with misvaluation it is possible that investors believe that the IC holds and believe that the managers will work hard even when they will not. In an equilibrium with misvaluation, investors may invest and find that the manager chooses to shirk. Therefore, shirking in equilibrium is possible. Thus, the IR constraint must ensure that the managers' decision to participate is rational even when their IC does not hold. We will return to this important innovation of our model later on.

Having derived all the relevant constraints of the model, we start by solving the optimization program that derives the maximum WTP of each type.

Proposition 1. The most a strategic acquirer is willing to pay, $\bar{V}^s$, is defined by

$$(4a) \quad R^s/p_H' - \gamma(\bar{V}^s + I - A_m)/p_H^2 = \max[B/p_H' \Delta p', \gamma A_m/p_H^2] \\ \text{if } \gamma A_m/p_H \geq B/\Delta p.$$

$$(4b) \quad R^s/p_H' - \gamma(\bar{V}^s + I - A_m)/p_H^2 = \max[B/p_H' \Delta p', (\gamma A_m - B)/p_H p_L] \\ \text{if } \gamma A_m/p_H < B/\Delta p.$$

The result of Proposition 1 is four equations that describe the highest potential WTP. Which equation is in effect depends on the model's parameters. In some cases, either the perceived IC or IR will bind at a point where the real IC is met (equation (4a)). Or the perceived IC or IR will bind at a point where the real IC is potentially not met (equation (4b)). In words, the strategic buyer is willing to raise its offer until either the perceived IC or an IR binds.

B. Private Equity Buyout

In this subsection, we consider a manager who combines forces with a PE investor as an alternative organizational form to manage the company's assets.

An optimal contract requires the manager to invest A_m and the uninformed investors to invest the balance of $\bar{V}^{\text{PE}} + I - A_m$. If the project fails, the contract then pays everyone nothing, if the project succeeds, the payoff R^{PE} is divided into $R_m^{\text{PE}} > 0$ for the manager, $R_{\text{PE}}^{\text{PE}} > 0$ for the PE investor, and $R_u^{\text{PE}} > 0$ for the uninformed investor, where $R_{\text{PE}}^{\text{PE}} + R_m^{\text{PE}} + R_u^{\text{PE}} = R^{\text{PE}}$. Investors will invest only if they believe the manager will work hard. However, if the PE firm monitors the manager (at a cost $c \geq 0$), investors need only believe that $p'_H R_m^{\text{PE}} \geq p'_L R_m^{\text{PE}} + b$. Therefore, the incentive-compatible investor belief requires that the manager be paid at least

$$(5) \quad R_m^{\text{PE}} \geq b/\Delta p',$$

and the belief that the PE monitors management requires the investors to believe that $p'_H R_{\text{PE}}^{\text{PE}} \geq p'_L R_{\text{PE}}^{\text{PE}} + c$. Therefore, the incentive-compatible investor belief also requires

$$(6) \quad R_{\text{PE}}^{\text{PE}} \geq c/\Delta p'.$$

If conditions (5) and (6) hold, then uninformed investors will invest $\bar{V}^{\text{PE}} + I - A_m$ as long as they also expect to earn γ on this investment. Thus, $p'_H R_u^{\text{PE}} = \gamma(\bar{V}^{\text{PE}} + I - A_m)$. Given the required return to investors, the manager earns $R_m^{\text{PE}} = R^{\text{PE}} - R_{\text{PE}}^{\text{PE}} - \gamma(\bar{V}^{\text{PE}} + I - A_m)/p'_H$ if the project is successful. Using this expression and the investors' perception of the PE's IC constraint, the investors think that the manager's IC constraint is

$$\begin{aligned} \text{Investor's view of IC: } R_m^{\text{PE}} &= \\ R^{\text{PE}} - c/\Delta p' - \gamma(\bar{V}^{\text{PE}} + I - A_m)/p'_H &\geq b/\Delta p'. \end{aligned}$$

However, uninformed investors potentially have an incorrect view. The manager will actually work hard (while the PE investor monitors) only if

$$(7) \quad R_m^{\text{PE}} = R^{\text{PE}} - c/\Delta p - \gamma(\bar{V}^{\text{PE}} + I - A_m)/p'_H \geq b/\Delta p.$$

This is the manager's true IC constraint.

In solving for the highest WTP, we can treat the manager and PE investor as one unit because they are both informed. Thus, either the manager will work hard and the PE investor will monitor or both will shirk.¹²

As before, the manager's expected return must also be greater than γA_m , otherwise the manager would rather invest A_m elsewhere. Therefore, the manager's individual rationality constraint is¹³

$$\text{Manager's IR} \quad \begin{cases} R_m^{\text{PE}} \geq \gamma A_m/p_H & \text{if manager's IC holds} \\ R_m^{\text{PE}} \geq (\gamma A_m - B)/p_L & \text{if manager's IC does not hold.} \end{cases}$$

Following the same logic as Proposition 1, the maximum WTP by a the PE type is contained in the following proposition:

¹²Specifically, we assume that $R_m^{\text{PE}} < B/\Delta p$, otherwise the manager behaves without monitoring.

¹³The PE IR always holds because if their IC holds then $R_{\text{PE}}^{\text{PE}} \geq c/\Delta p > c/p_H$ and if their IC does not hold than any positive $R_{\text{PE}}^{\text{PE}}$ is greater than or equal to the IR bound of 0.

Proposition 2. The highest WTP by a PE firm, $\bar{V}^{\text{PE}}$, is such that the following holds:

$$\begin{aligned} (8a) \quad R^{\text{PE}} - \gamma(\bar{V}^{\text{PE}} + I - A_m)/p'_H &= \max[(b+c)/\Delta p', \gamma A_m/p_H + c/\Delta p] \\ &\quad \text{if } \gamma A_m/p_H \geq b/\Delta p \\ (8b) \quad R^{\text{PE}} - \gamma(\bar{V}^{\text{PE}} + I - A_m)/p'_H &= \max[(b+c)/\Delta p', (\gamma A_m - B)/p_L] \\ &\quad \text{if } \gamma A_m/p_H < b/\Delta p. \end{aligned}$$

The same interaction between the perceived IC, the true IC, and the IR discussed previously for Proposition 1 explains that the maximum WTP is a step function with three expressions.

Although the point of the paper is to compare the PE offer to the strategic offer, the stand-alone firm will serve as a benchmark. The following corollary demonstrates the reservation price of a stand-alone firm:

Corollary 1. The reservation value of the stand-alone firm, $\bar{V}$, is such that the following holds:

$$\begin{aligned} (9a) \quad R - \gamma(\bar{V} + I - A_m)/p'_H &= \max[B/\Delta p', \gamma A_m/p_H] \\ &\quad \text{if } \gamma A_m/p_H \geq B/\Delta p \\ (9b) \quad R - \gamma(\bar{V} + I - A_m)/p'_H &= \max[B/\Delta p', (\gamma A_m - B)/p_L] \\ &\quad \text{if } \gamma A_m/p_H < B/\Delta p. \end{aligned}$$

One might have expected the stand-alone value of the firm to simply be the discounted net present value of the investment. Without misvaluation, the price would either be the net present value or would be constrained by the inability to borrow due to the moral hazard problem. With misvaluation, there is a difference between $B/\Delta p'$ and $B/\Delta p$ as well as $(b+c)/\Delta p'$ and $(b+c)/\Delta p$. Thus, it is the mispricing directly and through its interaction with the moral hazard problem that leads to differences in WTP. As we explained previously, mispricing also creates the possibility that managers, with or without a PE partner, shirk in equilibrium. This cannot happen in a standard model but is possible with mispricing.

This completes the characterization of both a strategic acquisition where two projects are organized under the same firm as well as a PE acquisition. Having examined the three possible organizational forms, we next compare them and derive the main results and predictions of the paper.

IV. The Coinsurance–Misvaluation Interaction Effect

In order to highlight the effect that results from debt and coinsurance, we first abstract away from the moral hazard part of our setup. We do so without loss of generality in the sense that this effect does not depend on the extent or existence of moral hazard between investors and management. The model contains moral hazard because moral hazard drives the need for a debt-like priority contract with investors. Given a debt contract, several coinsurance effects will arise. We will then need to go back and consider the effects of misvaluation on the moral hazard problem, which we will do in the next section.

To abstract away from the moral hazard effects, we simply need to limit the importance of the moral hazard problem until it does not affect prices. This requires only a small enough b and B , but for simplicity we assume that $b = B = 0$. We also assume $R^{\text{PE}} = R^s$, so there is both no moral hazard issue and no return benefit to any organizational form. The following proposition contains an important result of this paper:

Proposition 3. The Coinsurance–Misvaluation Effect. Absent moral hazard, debt misvaluation determines the relative WTP of PE and strategic acquirers. That is, $p'_H > p_H \Rightarrow \bar{V}^{\text{PE}} > \bar{V}^s$, $p'_H < p_H \Rightarrow \bar{V}^{\text{PE}} < \bar{V}^s$, and $p'_H = p_H \Rightarrow \bar{V}^{\text{PE}} = \bar{V}^s$. Moreover, the difference in WTP ($\bar{V}^{\text{PE}} - \bar{V}^s$) increases (decreases) with overvaluation (undervaluation).

Proof. First, using the results in Propositions 1 and 2 and the previous assumptions, the relevant equations are (4a) and (8a) since $A_m/p_H p_L \geq A_m/p_H^2 > 0$. The difference between $\bar{V}^{\text{PE}}$ and $\bar{V}^s$ is given by

$$\left(\frac{p'_H}{p_H}\right)\left(\frac{p'_H}{p_H} - 1\right)A_m.$$

The term is strictly positive if and only if $p'_H > p_H$, that is, when debt is overvalued. Moreover, $p'_H > p_H$ is a sufficient condition for the previous term to increase as p'_H increases. $\square$

The interaction of misvaluation and coinsurance helps PE firms to outbid strategic buyers. This occurs because the investors' misperception of the value of the projects causes them to also misperceive the value of coinsurance. The combination of projects effectively reduces the valuation mistake being made by investors. That is, they underestimate the default probability, but that makes them underestimate the value of the coinsurance benefit. The mistakes partially offset and thus an overvalued debt market cannot be as exploited by a strategic buyer as it can be by a stand-alone or PE buyer.

Without misvaluation, the coinsurance by itself does not influence the maximum WTP, because the effect is priced by the debt holders. We see this in the previous proposition because the difference between $\bar{V}^{\text{PE}}$ and $\bar{V}^s$ is 0 if there is no misvaluation ($p'_H = p_H$). Thus, it is the interaction between misvaluation and coinsurance that results in a differential WTP. This difference helps PE firms when the market is overvalued and helps strategic acquirers when the market is undervalued.

This section takes a fundamental difference between strategic and financial buyers (i.e., that strategic buyers have a current project) and shows how this interacts with misvaluation to create a differential WTP. It is a very intuitive result: If the projects are perceived to be very likely to pay off, then the value of any coinsurance is naturally perceived to be low, and vice versa if projects are perceived to be extremely risky. This suggests a potential mechanism for a changing ratio of strategic to financial acquirers that depends on the level of overvaluation in debt markets: When the perception of default is very low, the benefits to coinsurance will also be perceived as low, limiting the relative WTP of strategic acquirers.

A. Limiting Effects

A limiting force on the coinsurance effect occurs when the two firms' cash flows are positively correlated as opposed to independent, as assumed previously. In this case, the price effect shown in Proposition 3 is diminished and financial buyers enjoy a lower advantage compared to strategic buyers. An extreme example is the case of perfectly correlated projects. If the two projects are perfectly positively correlated, then the possibility of coinsurance disappears. The strategic acquirer scenario then becomes equal to the stand-alone case. The relevant comparison becomes the stand-alone and the PE cases, where the differences arise from the different agency costs and monitoring. We highlight this observation because it has empirical content:

Corollary 2. Strategic acquirers whose cash flows are more correlated with the target's are more able to outbid financial buyers when debt markets are overvalued, $\mu > 0$.

The model provides novel intuition as to why and when PE may dominate strategic buyers and thus how to examine the data. In the data, we would like to find at least suggestive evidence that relative PE activity is higher in periods when debt is misvalued. Moreover, if the coinsurance effect is correct, then Corollary 2 implies that the relative dominance of PE should be even greater as compared to conglomerate strategic buyers (buyers from different industries). This is because conglomerate buyers are more diversified and should be more impacted by the coinsurance/misvaluation interaction effect. These are direct empirical implications of the coinsurance effect in the model. The next section develops the moral hazard effects and the empirical tests for how moral hazard interacts with misvaluation.

V. Moral Hazard and Misvaluation

In order to analyze how the moral hazard problem and its corporate governance implications are affected by the possibility of misvaluation, we will start by adding some additional structure to our existing model. In particular, we allow p'_H and p'_L to be functions of an underlying parameter that measures the extent of asymmetric information or misvaluation, μ . That is, with a slight abuse of notation, let us define $p'_H \equiv p'_H(\mu)$, where p'_H is a continuous, differentiable, and strictly increasing function of μ over the real domain, $p'_H: \mathbb{R} \rightarrow [0, 1]$.¹⁴ Moreover, in the absence of misvaluation ($\mu = 0$), the perceived probability coincides with the true probability, $p'_H(0) = p_H$; since $p' \in [0, 1]$, we also require that $\lim_{\mu \rightarrow \infty} p'(\mu) = 1$ and $\lim_{\mu \rightarrow -\infty} p'(\mu) = 0$. Given this structure, $\mu > 0$ results in overvaluation, while $\mu < 0$ results in undervaluation.¹⁵

¹⁴Therefore, $p'_L(\mu) = \theta p'_H(\mu)$.

¹⁵We note also that although the effects in the previous section arose because the security issued by the firm was debt, in this section the effects are due to the underlying asset misvaluation and not to the type of security. In our model, moral hazard resulted in the use of a debt security, which then led to the previously discussed effects. However, the moral hazard effects should impact a firm if they become misvalued regardless of whether they used debt.

In order to understand the potential impact of overvaluation, it is important to first establish its effect on the moral hazard problem. The next proposition does so.

Proposition 4. Overvaluation reduces the perceived moral hazard cost, but makes equilibrium shirking possible.

A striking result occurs when overvaluation is introduced into a standard moral hazard setting. Because misvaluation causes the investor to incorrectly perceive probabilities, a difference between the true and perceived incentive compatibility constraint appears. Absent misvaluation, only the no-private-benefit project is funded in equilibrium because it is the only economically viable project. However, with misvaluation, if the perceived IC holds while the true IC does not, then a firm will be funded even though the manager shirks in equilibrium.

Overvaluation decreases the perception of the moral hazard problem and increases investors' willingness to lend relative to the case of $\mu = 0$. But actual misbehavior is possible even though the investor perception is that the problem has gotten better. In fact, equilibrium shirking is possible precisely because investors perceive the moral hazard problem to be better than it actually is. This result is a significant departure from the classical moral hazard paradigm.

Some previous work has suggested that overvaluation positively impacts investment (see Morck, Shleifer, and Vishny (1990), Stein (1996), Baker, Stein, and Wurgler (2003b), Gilchrist, Himmelberg, and Huberman (2005), Polk and Sapienza (2009), and Campello and Graham (2013)) and correlates with shirking and other misbehavior by managers (see Bertrand and Mullainathan (2001), Povel, Singh, and Winton (2007), and Wang, Winton, and Yu (2010)).¹⁶ Our work is closest to the ideas of Jensen (2005), Bolton, Scheinkman, and Xiong (2005), and Bolton, Scheinkman, and Xiong (2006), wherein overvaluation worsens the moral hazard problem. We build on all of the previous work by both considering overvalued debt and also considering the effects of better governance. Furthermore, we apply these ideas to better understand takeover activity.

A. The Monitoring Effect

One of the main differences between financial and strategic acquirers is that the monitoring capacity of PE sponsors alleviates the agency cost caused by moral hazard. In order to highlight the effect that results from the interaction between overvaluation and moral hazard we assume the moral hazard problem is large enough that it affects prices. Specifically, in order to isolate the impact of misvaluation on moral hazard we assume that private benefits are large enough relative to the firm's cash (i.e., $b/\Delta p > \gamma A_m/p_H$). This assumption is without loss of generality in the sense that the moral hazard effect does not depend on the existence or magnitude of the issues that stem from diversification/coinsurance. We also now allow R^{PE} and R^s to differ from each other and consider the set of synergies and PE value enhancements that result in one or the other type being willing to

¹⁶Bertrand and Mullainathan (2001) suggest that chief executive officers tend to "skim" profits in good times. Povel et al. (2007) and Wang et al. (2010) show that investor monitoring tends to fail to prevent fraud when the perceived industry conditions are good. This work all suggests governance and regulatory oversight can be particularly important during booms.

pay more. The following proposition demonstrates the monitoring effect, which is another central point in our paper, and shows that overvaluation both alters the set of financial buyers who are willing to pay more than a strategic acquirer and also changes the differential offer amount:

Proposition 5. The Monitoring Effect. If debt is overvalued ($\mu > 0$), then for any given R^s , increasing μ increases the set of R^{PE} values such that $\bar{V}^{\text{PE}} > \bar{V}^s$. In other words, if debt is overvalued, the set of parameter values such that the financial buyer is willing to pay more than the strategic buyer increases with the extent of overvaluation. Furthermore, within this set of financial and strategic buyers, overvaluation increases the financial buyer's relative WTP, $\partial(\bar{V}^{\text{PE}} - \bar{V}^s)/\partial\mu > 0$.

The perception of the agency cost changes the value of monitoring and hence alters the ability of the firm to borrow money, which in turn alters the WTP. Because with overvaluation the perception of the moral hazard problem is better (Proposition 4), it implies that investors perceive the monitoring costs to be lower and thus perceive the total value creation of a financial buyer to be greater. This increases the investors' willingness-to-lend to the financial buyer, which in turn both changes the set of PE firms that can win and increases their relative WTP.

VI. The Extensive Margin: Aggregate Acquisition Activity

The previous sections analyzed the relative ability of the different organizational forms to create value and thus bid higher for a given target. We separated the main drivers of WTP into the coinsurance effect and the monitoring effect. These were valuation, or "price," effects at the intensive margin. We now focus on the predictions of the model that relate to the aggregate number of firms that we expect could be acquired under one or another organizational form. We call this the extensive margin. We ask when overvaluation allows more companies to raise financing under the sponsorship of a PE firm versus a strategic deal. This extensive margin together with the price effects discussed in the previous sections provide a complete picture of the determinants of acquisition activity.

The agency problem creates an ex ante financing constraint and establishes a minimum amount of internal funds needed to obtain outside financing. This cutoff determines whether a particular organizational form (PE, strategic, or stand-alone) will be able to sell a debt instrument in order to finance the investment or acquisition. As this cutoff changes, the number of potential PE or strategic buyers will increase or decrease, further explaining the shifting type of acquirer (financial or strategic).

We start by solving for the minimum required cash for each type of firm organization.

The minimum cash is always the point at which the perceived pledgeable income is equal to the amount raised from outsiders, $I - A_m$. For a PE acquirer, the minimum acceptable cash is

$$(10) \quad A_m \geq \bar{A}_m^{\text{PE}} = I - \frac{p'_H}{\gamma} \left(R^{\text{PE}} - \frac{b+c}{\Delta p'} \right).$$

For a strategic acquisition, the minimum A_m is defined by

$$(11) \quad A_m \geq \bar{A}_m^s = I - \frac{p'_H}{\gamma} \left(R^s - \frac{B}{\Delta p'} \right).$$

And finally, for a stand-alone firm,

$$(12) \quad A_m \geq \bar{A}_m = I - \frac{p'_H}{\gamma} \left(R - \frac{B}{\Delta p'} \right).$$

Taken together, equations (10)–(12) imply that when $\bar{A}_m^{\text{PE}} < \bar{A}_m^s$, firms with $A_m \in [\bar{A}_m^{\text{PE}}, \bar{A}_m^s)$ can be acquired only by a financial sponsor. Moreover, because such a mass of firms possesses the smallest amount of internal funds, they represent the highest leverage deals.

The following proposition and corollary compare the three cutoffs:

Proposition 6. PE Buyout Activity and Strategic Activity. Debt overvaluation:

- i) loosens financial constraints on all organizational forms, allowing the funding of deals with lower internal funds; and
- ii) increases the set of parameter values for which the minimum acceptable cash in a PE deal is less than in a strategic deal ($\bar{A}_m^{\text{PE}} < \bar{A}_m^s$). In other words, for any given R^s , increasing μ increases the set of R^{PE} such that $\bar{A}_m^{\text{PE}} < \bar{A}_m^s$.

The driver of point i) stems from the same intuition as Proposition 4: Overvaluation reduces the perception of the moral hazard problem, therefore loosening the constraints on all types of organizational forms. But more important for the purposes of our paper, we find that overvaluation unambiguously increases PE activity because it increases the situations (the set of model parameters) in which only a PE firm will be able to acquire the target. The following corollary further demonstrates that overvaluation also increases overall acquisition activity by both financial and strategic acquirers:

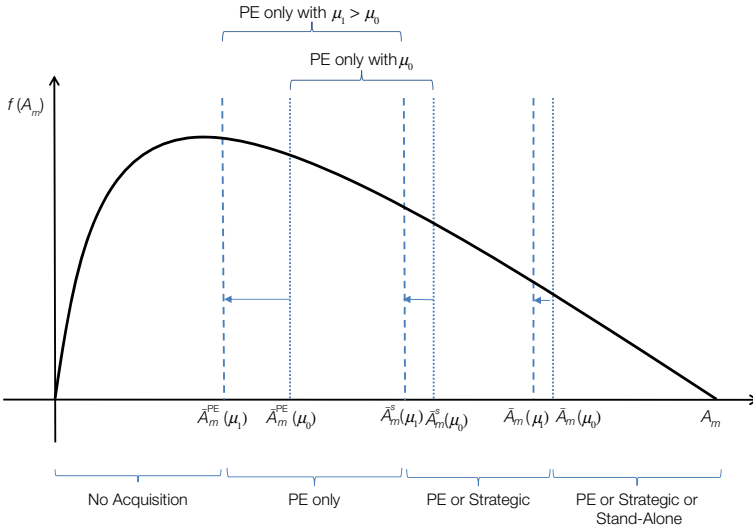
Corollary 3. Provided $R^{\text{PE}} > R$ and $R^s > R$, overvaluation increases both $(\bar{A}_m - \bar{A}_m^{\text{PE}})$ and $(\bar{A}_m - \bar{A}_m^s)$.

This corollary tells us that overvaluation increases the set of firms that must be acquired by either a financial or a strategic buyer, with the only requirement being that synergies must be positive and PE must increase stand-alone firm profits (or else no firms would be acquired). In summary, the overvaluation's extensive margin not only changes the ratio of financial to strategic acquirers but also increases overall acquisition activity.

We have shown that with overvaluation there exists an increase in the mass of firms with $A_m \in [\bar{A}_m^{\text{PE}}, \bar{A}_m^s)$, which only financial bidders can buy and hence for which a PE buyout is the optimal organizational form. For firms such that $A_m \geq \bar{A}_m^s$, both types of buyers are likely to be present in an auction for the company. And furthermore, if we believe that some strategic buyers have synergies greater than or equal to 0, then $\bar{A}_m \geq \bar{A}_m^s$. For firms with enough capital, all three organizational forms are possible. Figure 2 illustrates the case that becomes more

FIGURE 2
Effect of Overvaluation on Minimum Required Capital

Figure 2 plots, for a given distribution of target firms given their amount of cash, $f(A_m)$, the minimum acceptable cash, $\bar{A}_m$, under each organizational structure. It corresponds to the case of $\bar{A}_m > \bar{A}_m^s > \bar{A}_m^{\text{PE}}$. In particular, it illustrates that as overvaluation increases from μ_0 to μ_1 , the area where only financial buyers can finance the deal increases, as explained in Proposition 6 and Corollary 3.



likely with overvaluation, $\bar{A}_m > \bar{A}_m^s > \bar{A}_m^{\text{PE}}$. Thus, with overvaluation, PE firms can purchase targets with even less cash and take on even greater leverage. Figure 2 illustrates this result on a given distribution of firms based on A_m , $f(A_m)$.

Combining the results from this section with the previous results on pricing, we have that leverage should be higher in PE deals for two reasons. First, when overvaluation causes financial buyers to pay more, they finance this with increased leverage. Second, since I is fixed, if a PE buyer alleviates the financing constraint, those firms that now access investors will do so with larger amounts of debt.

This is consistent with the empirical evidence that PE deals are not only highly leveraged but also have leverage that depends on debt market conditions more than fundamental factors (see Axelson et al. (2013)). However, Axelson et al. (2013) argue that because leverage and returns are negatively related, their findings are inconsistent with a market-timing story. Our model shows that overvaluation may cause WTP to increase, but this does not imply increased returns. The benefit of overvalued debt markets may accrue to the target.

Overall, we have shown how debt overvaluation can have an effect both on the acquirer WTP and on the acquirer ability to finance the deal that could explain why we see increased financial sponsor activity that correlates with overvaluation in the debt market, particularly in highly active debt markets.

VII. Robustness and Discussion

This section discusses the implications of extending the model in several interesting dimensions. First, we consider a PE buyer with her own capital. Then

we introduce updating by investors. Finally, we discuss divestitures instead of acquisitions. The section ends with a qualitative discussion of the most current LBO wave through the lenses of our theory.

A. The Impact of Cash

In order to more easily compare the organizational forms, we assumed previously that PE investors brought no additional cash to the transaction. In this subsection, we relax this assumption. If the financial buyer brings cash to the acquisition, since monitoring ability is scarce, the financial buyer commands a premium return on his cash, $\gamma_{PE} \geq \gamma$.¹⁷ Therefore, we endogenize the amount of cash brought by the financial buyer by finding the amount that allows for the highest offer. Thus, the PE's individual rationality constraint will determine PE capital, A_{PE} . If the PE IC holds, then the PE investor is willing to alter his capital until the point at which $\gamma_{PE} A_{PE} = p_H c / \Delta p$. Thus, investors must perceive that the PE investor will monitor and it must be individually rational for him to do so. If the PE IC does not hold, then the PE investor does not invest any capital.

Proposition 7. Allowing financial buyers to have cash increases their WTP but does not change the effect of overvaluation on their WTP.

Thus, if we endogenize the financial buyer's cash, we see that this provides another reason that the financial buyer may offer a higher price, but cash does not change the effects of misvaluation.

B. Misvaluation and Updating

Given the richness of the model, we do not examine how uninformed investors might update their valuation estimates given the actions of the informed. The ideas we present here would continue to hold with updating as long there was enough noise in the model that informed actions were not fully revealing. For example, in Rhodes-Kropf and Viswanathan (2004), the acquirer stocks are mispriced but sellers cannot distinguish between overvaluation or high synergies, so when market wide overvaluation is high sellers update prices but still overestimate the synergies. In our case, the probabilities of success and the error estimates might arise from many different sources. If informed players knew the mistake but not the components of the mistake then actions by the informed could not be fully revealing. By not considering updating, essentially we are assuming that there is enough noise in a single deal that investors would learn little about aggregate debt market mispricing from the actions of players in one particular deal. The inclusion of updating could create wave dynamics, as in Rhodes-Kropf and Viswanathan, in that each deal would cause small shifts in the market-wide misvaluation until early deals were realized and revealed the truth, ending the misvaluation.

C. Divestitures and Asset Sales

Even though we have motivated this paper in the context of acquisition activity, its predictions and implications go beyond asset expansion and can be more generally related to overall restructuring activity. One example of this broader

¹⁷The minimum acceptable rate of return when the PE IC holds is determined by the condition $p_H c / \Delta p - c = \gamma A_{PE} = \gamma p_H c / \gamma_{PE} \Delta p$, which translates into $\gamma_{PE} = \gamma p_H / p_L$.

interpretation can be made in the context of optimal asset sale policies. When debt is overvalued, a diversified company can potentially unlock value by divesting a division. As a stand-alone entity the division should be better able to extract information rents from lesser informed investors. However, a divestiture will be an optimal strategy only if, of course, there are no significant synergies between the division to be divested and the rest of the divisions that comprise the original firm. Hence, in terms of overvalued debt markets, our article suggests not only more acquisition activity (with potentially more financial buyers) but also more divestitures or asset sales undertaken by diversified companies or conglomerates. The reason for this prediction is the same as that driving our acquisition results, namely, the interaction of information asymmetries and diversification of cash flows.

D. The Merger Wave of 2005–2007

The starting point of this article was the observation that the one thing that seemed to characterize the last wave of acquisition activity in 2005–2007 was the more predominant role of financial buyers relative to strategic buyers. It has been argued by both industry practitioners and some academics that this period was characterized by a period of potentially overvalued debt and hence “too low” yields and increased lending. This casual observation is consistent with, and predicted by, our model. Our theory predicts increased acquisition and lending activity during periods of overvalued debt as well as a relative preponderance of financial buyers vis-à-vis strategic buyers.

VIII. Evidence

Although this is predominantly a theory article, we present some evidence that supports the idea that debt market misvaluation may be an important driver of the shifting importance of financial buyers over strategic buyers. We hope this will encourage future work to take a more in-depth look.

The impact of bond market misvaluation has been the object of relatively few studies compared to equity misvaluation. Greenwood and Hanson (2013) investigate overheating in the credit market and show that when aggregate credit increases, the average quality of issuers deteriorates; following periods of poor issuer quality, corporate bonds significantly underperform U.S. Treasury bonds. Furthermore, Baker, Greenwood, and Wurgler (2003a) find that when firms issue more long-term debt, future excess bond returns are low.

In practice, debt misvaluation would imply that firms with relatively high ratings default too often. We exploit this implication in order to measure misvaluation. As it turns out, Moody's tracks the ex post accuracy of its rating measures, using the so-called average position (AP). According to Moody's, the position of any debt issuance is defined as the share of debt issuances in a cohort rated better than it in the year the debt was issued. It assumes each debt issuance occupies the midpoint of its rating category. For example, the position of every Aa2 debt issuance is the share of the cohort rated Aaa or Aa1 plus half the share rated Aa2.

The 5-year AP is then calculated simply as the average of the positions of the debt issuances that defaulted within 5 years. Intuitively, a more powerful rating

system should have low-rated defaults and high-rated nondefaulters, meaning the AP should be high.

A higher AP reflects better *ex post* accuracy of Moody's ratings. If all defaulters were initially given the lowest rating, then the AP would approach 1. Alternatively, if all defaulters were initially given a random rating, then the AP would be about $\frac{1}{2}$. And if all defaulters were initially given the best rating, then the AP would approach 0. Thus, if more defaulters were given higher ratings, the AP would fall. Therefore, our concept of overvaluation (higher-rated companies defaulting too often) will be captured by lower average positions. Potentially, Moody's inaccuracies may be perfectly understood and priced by bond market participants. Thus, we cannot be certain that AP captures debt overvaluation. However, there is support for the idea that credit ratings affect bond prices (see, e.g., Campbell and Taksler (2003)).

Moody's has provided us with quarterly cohort 5-year AP measures; using the raw data from Moody's, we have constructed AP for each of the Fama–French 12 industries. We also use an alternative measure of misvaluation: the ratio of ratings downgrades to upgrades 3 years after the year of interest. We get similar inferences using this alternative. In head-to-head tests, the downgrade-to-upgrade ratio loses significance in the presence of AP.¹⁸ Therefore, we report the results from the use of AP.

We control for other measures of bond market conditions and activity, such as the 5-year U.S. Treasury rate, the high-yield credit spread, and the average spread over the federal funds rate for commercial and industrial loans. As a general measure of equity valuations, we also control for the median market-to-book ratio (based on the Compustat population) and the standard deviation of the market-to-book ratio. For the level of economic activity, we employ change in gross domestic product (GDP).

Our measure of PE activity is the fraction of the value of all deals for public targets accounted for by financial sponsors (PE or leveraged buyouts), which we calculate on an annual basis. Figure 3 plots the average position and the percentage of PE activity. The dependent variable is the fraction of value of all deals done by PE in a given Fama–French industry. Alternatively, the dependent variable is an indicator variable set to 1 for high PE periods, defined as years in which the average fraction of activity accounted for by PE activity is more than 8.86% (which is the sample mean of the value ratio of PE deals).

Since our predictions are sharper for the relative impacts of misvaluation on PE versus conglomerate activity, we also construct the PE/(PE + CONGLOMERATE) ratio using the value of PE and conglomerate deals. Because we are interested in the coinsurance effect we define a conglomerate acquirer as an acquirer that operates in Fama–French industries other than the target company does; these are clearly diversified entities.¹⁹

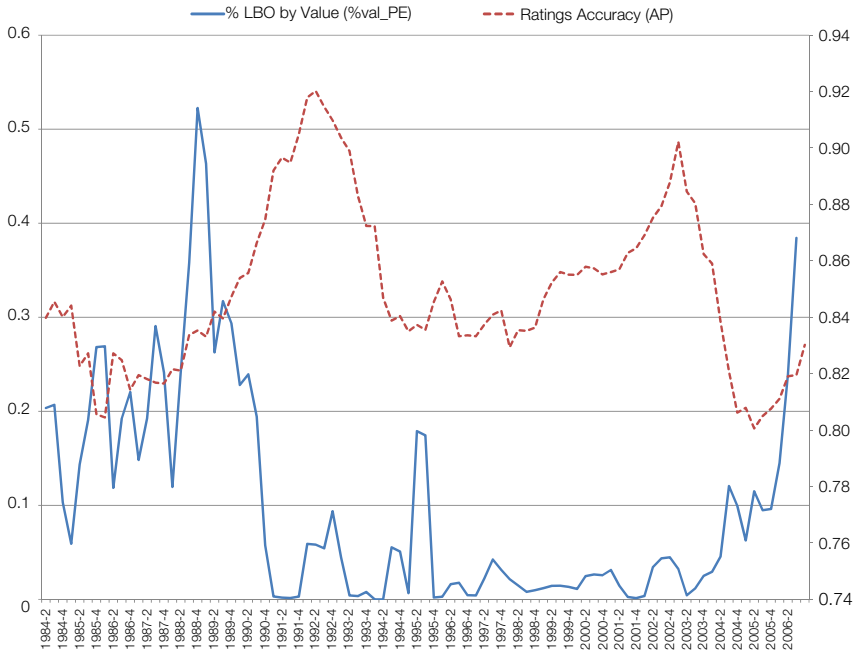
Table 2 presents summary statistics for the variables used in our estimation. The average fraction of activity accounted for by PE acquirers is 8.86%, and

¹⁸We suspect this is because the downgrades-to-upgrades ratio is not cohort specific, making it difficult to identify exactly when the misvaluation occurred.

¹⁹We employ the Fama–French 12-industry classification.

FIGURE 3
Moody's Average Position (AP) and Private Equity (PE) Activity (%val_PE)

Figure 3 plots Moody's average position (AP) and the share of private equity (PE) mergers and acquisitions (M&A) activity (right axis and left axis, respectively), by quarter from 1984–2006. Moody's AP was provided by Moody's. The share of PE activity was calculated based on the value of all M&A transactions involving a public target with transaction values greater than \$10 million and lower than \$1 billion, recorded in the SDC Platinum database.



27.27% of our years are high PE periods. The $PE/(PE + CONGLOMERATE)$ ratio averages 21%. The aggregate average position fluctuates within a relatively tight range (between 81% and 92%), so even relatively small changes in the average position reflect large changes in bond market “misvaluation.” At the industry level, on the other hand, the range expands to 50%–99%.

We start by estimating a Tobit model of the fraction of financial buyers. We use a Tobit in order to account for the potential censoring of the dependent variable at 0. We present the results as marginal effects in Table 3 (columns 1–6), where the first row contains aggregate average position (AP), our measure of bond market pricing accuracy, and the second row the same variable calculated at the Fama–French 12-industry level, AP_FF12 . Initially we run regressions without average position so that we see the expected effects for the control variables. For example, the high-yield spread is negatively correlated with PE activity. We then add average position. It is significantly negative, confirming that in periods when “overvaluation” is high, as measured with a low AP, PE activity increases as a fraction of total M&A activity. The effect is economically large as well: A 1% decrease in aggregate AP increases the fraction of PE activity by 3.2%. For industry-level AP, the range of values is about 5 times that of the aggregate AP and the coefficient about 1/5th of the aggregate AP, so that a 1% decrease

TABLE 2
Summary Statistics

In Table 2, VALUE_BY_PE is the sum of the value of transactions by private equity (PE) sponsors divided by the total mergers and acquisitions (M&A) value. This is calculated for each Fama-French 12-industry classification. HIGH_PE is a dummy variable that takes the value of 1 if PE participation as a percentage of total deal value is above 8.86%, and 0 otherwise. AGGREGATE_AP is Moody's 5-year average position, a measure of the accuracy of a given year's bond ratings over the following 5 years. FF12_AP is the same measure at the industry level. We define average position (AP) more formally in Section VIII. 5_YEAR_TREASURY is the yield to maturity for 5-year U.S. Treasury bonds. The HIGH_YIELD_SPREAD is the difference between the Bank of America Merrill Lynch High-Yield 100 Index yield and the 5-year U.S. Treasury yield. The C&I_SPREAD is the spread between the average rate on commercial and industrial loans and the federal funds rate (series E.2 from the Federal Reserve). The CHANGE_IN_GDP is the annual growth rate of gross domestic product (GDP). Median M/B and Std. dev. M/B are the median and standard deviation of the market-to-book ratio for Compustat firms (data are winsorized at the 1st and 99th percentiles) at the Fama-French 12-industry level. There are 176 annual industry observations from 1984 to 2005.

Variable	Mean	Std. Dev.	Min.	Max.
VALUE_BY_PE	0.0886	0.1579	0.0000	0.8374
HIGH_PE	0.2727	0.4466	0.0000	1.0000
AGGREGATE_AP	0.8540	0.0299	0.8079	0.9179
FF12_AP	0.8288	0.0920	0.5000	0.9935
5_YEAR_TREASURY	5.9028	1.6809	3.0300	9.0900
HIGH_YIELD_SPREAD	6.0638	2.2299	3.7710	12.4760
C&I_SPREAD	1.6216	0.2461	1.1900	2.1200
CHANGE_IN_GDP	3.0833	1.2591	-0.2330	4.8254
Median M/B	1.5301	0.4352	0.9024	3.2877
Std. dev. M/B	9.5218	20.3417	0.2864	158.0153
No. of obs.	176			

TABLE 3
Financial Buyer Activity Relative to Strategic Buyer Activity

Table 3 shows Tobit regressions (columns 1–5) and a logistic regression (column 6). VALUE_PE is the fraction of total private equity (PE) participation over total deal value. HIGH_PE is a dummy variable that takes the value of 1 if PE participation as a percentage of total deal value is above 8.86%, and 0 otherwise. Observations are at the annual and industry level. Reported values are marginal effects. Standard errors are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Variables	VALUE_PE					HIGH_PE
	1	2	3	4	5	6
AGGREGATE_AP			-3.2256*** (0.8887)			
FF12_AP				-0.6240*** (0.1740)	-0.5318*** (0.2047)	-0.7756** (0.3449)
5_YEAR_TREASURY	0.0429*** (0.0119)	0.0413*** (0.0122)	0.0460*** (0.0116)	0.0448*** (0.0117)	0.0389*** (0.0114)	0.0888*** (0.0207)
HIGH_YIELD_SPREAD	-0.0192** (0.0079)	-0.0209*** (0.0079)	0.0135 (0.0120)	-0.0126 (0.0079)	-0.0158** (0.0078)	-0.0158 (0.0151)
C&I_SPREAD	0.2942*** (0.0905)	0.2801*** (0.0921)	0.2314*** (0.0877)	0.2733*** (0.0879)	0.2356*** (0.0843)	0.4609*** (0.1763)
CHANGE_IN_GDP	0.0217 (0.0152)	0.0246 (0.0150)	0.0118 (0.0146)	0.0205 (0.0143)	0.0206 (0.0140)	0.0229 (0.0283)
M/B (industry)		-0.0810** (0.0383)	-0.0816** (0.0365)	-0.0704* (0.0366)	-0.1187* (0.0641)	-0.1097 (0.0848)
Std. dev. M/B		0.0007 (0.0009)	0.0005 (0.0008)	0.0009 (0.0008)	0.0011 (0.0009)	0.0020 (0.0015)
Industry fixed effects	No	No	No	No	Yes	No
No. of obs.	176	176	176	176	176	176

in industry AP increases the ratio of PE activity by 0.6%. Since the mean of PE activity is 8.9%, it is indeed economically important. Interestingly, the high-yield spread loses significance when debt mispricing (AP) is included (see columns 3, 4,

and 6). Thus, high PE activity does not seem to be just about changing economic conditions but rather is highly related to ratings mistakes.²⁰

Column 5 of Table 3 includes industry fixed effects to control for industry unobserved heterogeneity. Note that the coefficient on AP continues to be significant, even though, as we would expect, it is slightly less negative than in column 4. Thus, when an industry's AP is above its average, PE activity is also above its industry average. The smaller coefficient in column 5 suggests that some of the effect in column 4 comes from cross-industry effects: Industries with lower AP tend to have higher PE activity than those with higher AP.

For robustness, column 6 in Table 3 presents logistic regressions predicting the probability of high PE activity. The effect of AP has the predicted sign and is significant. In unreported results, we also rerun our analysis with PE activity measured as the fraction of the count (rather than value) of all deals for public targets accounted for by financial sponsors and find similar results. Finally, in untabulated results, we include the aggregate dollar amount raised by PE firms as a control (as well as lagged dollar amount raised) and find similar coefficients. This controls for concerns that the effect of PE dominance may come from increased capital availability rather than the shifting bond market misvaluation.

In Table 4 we continue to test predictions of our theory. First, an implication of the coinsurance effect, shown in Corollary 2, predicts that a conglomerate acquirer should be even more impacted than a nonconglomerate acquirer in their relative ability to acquire when debt markets become overvalued because they are diversified and most impacted by the coinsurance effect. In Table 4 (column 1 and 2) we present Tobit regressions, the dependent variable of which is the ratio of PE activity to PE plus conglomerate activity, $PE/(PE + CONGLOMERATE)$. AP is again significantly negative, confirming that in periods when misvaluation increases, as measured by low AP, PE activity increases relative to conglomerate activity. The effect is economically large as well: A 1% decrease in AP increases the fraction $PE/(PE + CONGLOMERATE)$ by 4.83%. Since the mean of $PE/(PE + CONGLOMERATE)$ is 24.2%, it is indeed economically important. As predicted by Corollary 2, the effect is 50% larger than it is for all acquirers (see column 3 of Table 3). Thus, the relative impact of PE to conglomerates is even stronger than its effect on overall activity, as predicted by our theory. This finding is supportive of the coinsurance effect and is neither an obvious prediction or a previously known fact.

Columns 3 and 4 of Table 4 reflect a prediction of the monitoring effect. Proposition 6 and Corollary 3 predict that when the parameters are such that the monitoring effect is relevant, misvaluation causes an increase in debt issuance. When we see higher debt issuance, we should also expect the impact of misvaluation on PE's ability to acquire a target to be even larger. Thus, we add debt issuance to the regression and interact it with AP. We define $HIGH_ΔD$ as a year that belongs to the upper quartile of the aggregate change in debt issuance for all Compustat firms in any given year. The relevant term for the purpose of this

²⁰Our treasury results are consistent with the prediction of Haddad et al. (2011), who argue that more LBOs should occur when risk-free rates are high. It is not surprising that PE activity is related both to changing economic conditions as well as misvaluation.

TABLE 4
Financial Buyer's Activity: Further Tests

Table 4 shows Tobit regressions. VALUE_PE is the fraction of total private equity (PE) participation over total deal value. PE_CONG is defined as PE/(PE+ CONGLOMERATE) and analogous to VALUE_PE, except that only deals by PE sponsors and by conglomerate acquirers are included. Conglomerate acquirers are defined as acquirers with Standard Industrial Classification (SIC) codes different from the target firm according to the Fama-French 12-industry classification. HIGH_ΔD is an indicator that is 1 in a year that belongs to the upper quartile of the aggregate change in debt issuance for all Compustat firms in any given year. Observations are at the annual level and either at the industry or aggregate level. Reported values are marginal effects. Standard errors are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Variables	PE_CONG		VALUE_PE	
	1	2	3	4
AGGREGATE_AP	-4.830*** (1.545)	-4.264*** (1.474)		-3.1736*** (0.8993)
FF12_AP			-0.4789** (0.1972)	
HIGH_ΔD			0.6107* (0.3471)	4.6679* (2.5304)
AP × HIGH_ΔD				-5.4247* (2.9672)
FF12_AP × HIGH_ΔD			-0.7786* (0.4339)	
5_YEAR_TREASURY	0.052*** (0.020)	0.040** (0.011)	0.0445*** (0.0117)	0.0322** (0.0134)
HIGH_YIELD_SPREAD	0.0100 (0.0212)	0.0003 (0.0207)	-0.0141* (0.0079)	0.0182 (0.0123)
C&I_SPREAD	0.438*** (0.139)	0.349** (0.136)	0.2735*** (0.0883)	0.1511 (0.0935)
CHANGE_IN_GDP			0.0198 (0.0142)	-0.0037 (0.0162)
M/B (industry)	-0.0942 (0.0644)	-0.216* (0.112)	-0.0733** (0.0363)	-0.0719** (0.0362)
Std. dev. M/B	-0.0003 (0.0015)	0.0005 (0.0017)	0.0009 (0.0008)	0.0006 (0.0008)
Industry fixed effects	No	Yes	No	No
No. of obs.	173	173	176	176

analysis is the interaction between the indicator for large debt issuance times and AP (both aggregate and industry level). Our theory would be consistent with the interaction term being negative. As predicted, the interaction has a significantly negative sign and debt overvaluation and debt issuance are positively correlated, providing indirect evidence of overvaluation and the impact of the monitoring effect on PE activity.

We find results that are consistent with the predictions of our theory and that are not obvious ex ante. Our proxy for debt market misvaluation correlates positively with the ratio of financial to strategic buyers, even after controlling for other relevant factors. Furthermore, the effect is stronger if the ratio is against strategic conglomerates, as predicted by the theory. While we must strongly qualify our results because we cannot be sure we have captured misvaluation, we include these results to provide support for the relevance of a theory of misvalued debt and PE activity. We expect future work to empirically explore this idea in greater detail.

IX. Concluding Remarks

In this article, we introduce debt market misvaluation in M&A. We also highlight and explain the oscillating pattern of financial versus strategic acquirers within merger waves.

We demonstrate that debt overvaluation can explain the increased activity and relative dominance of financial buyers. This is not obvious, because an overvalued debt market should raise the value of stand-alone firms as well as the WTP of both financial and strategic acquirers, as they can all access cheap debt. We model how misvaluation interacts with both coinsurance (Lewellen (1971)) and the moral hazard problem, to give financial firms a relative advantage when debt markets are overvalued.

We find a strong correlation in the data between measures of debt market misvaluation and the fraction of acquisition activity due to PE acquirers. This is consistent with our theory, which provides a possible explanation for this correlation. Further predictions of our theory are supported by the data. Thus, the evidence is consistent with the idea that misvaluation in the debt market drives the relative dominance of PE activity.

Overall, by combining moral hazard and coinsurance with the idea of debt misvaluation, we gain considerable insight into a previously unexplored pattern. We hope that future work will further examine the impact of potentially misvalued debt markets and show its relevance to M&A activity in the same way that so much has followed from the ideas of equity misvaluation.

Supplementary Material

Supplementary Material for this article is available at <https://doi.org/10.1017/S0022109019000139>.

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